

FUNDAMENTAL ECONOMIC PROBLEMS

The central economic problem for society is how to reconcile the conflict between people's limitless desires for goods and services, and the scarcity of resources (labour, machinery, and raw materials) with which these goods and services can be produced.

There are four basic problems of economic facing any society, these are:

- i. What to produce.
- ii. How to produce.
- iii. For whom to produce and
- iv. Efficient use of resources.

Basically, economic problems of society arise because of scarcity of resources. In answering the questions 'what, how and for whom to produce' economics explain how scarce resources are allocated between competing claims on their use.

Firms, households and government are faced with the problem of deciding what type of goods to produce. Remember, the resources of society are limited. If more resources are used to produce one kind of goods, there will be less for the other. The types of goods and services which will be produced depend among other things on the needs of the society and the available human and material resources. In a capitalist or free enterprise

Scarcity

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2. economy, the price system will determine the type and quantity of goods and services that should be produced.

How to Produce? This deals with the method to be used in producing goods and services. Methods of production may be labour intensive or capital intensive. Labour intensive method involves the use of more workers with less machinery while capital intensive use more machinery and less labour. Here, machines will be used to replace labour.

For whom to produce? Who will enjoy the goods produced and what quantity he will enjoy? In a free enterprise economy, the type of goods and the quantity of goods a person enjoys will depend on his income or purchasing power. Producers will always direct their production efforts toward those sections of the society where there is higher demand which will enable them to maximize profits.

DEMAND AND SUPPLY

Demand refers to the quantity of a commodity which consumers are willing to pay at a particular price over a period of time. Demand relates to the behaviour of buyers. Supply on the other hand refers to the quantity of goods or commodity which sellers are willing to sell at a particular over a period of time. Price and qty cannot be considered separately. Price determines one willing to buy and sellers are willing to sell.

Demand Schedule - It shows the relationship between the various quantities of commodity that will be bought at different prices. Below is an example of demand schedule.

Price of bread	Qty demanded
1200	5
1000	10
800	15
600	20
400	25
200	30

We can represent the demand schedule graphically.

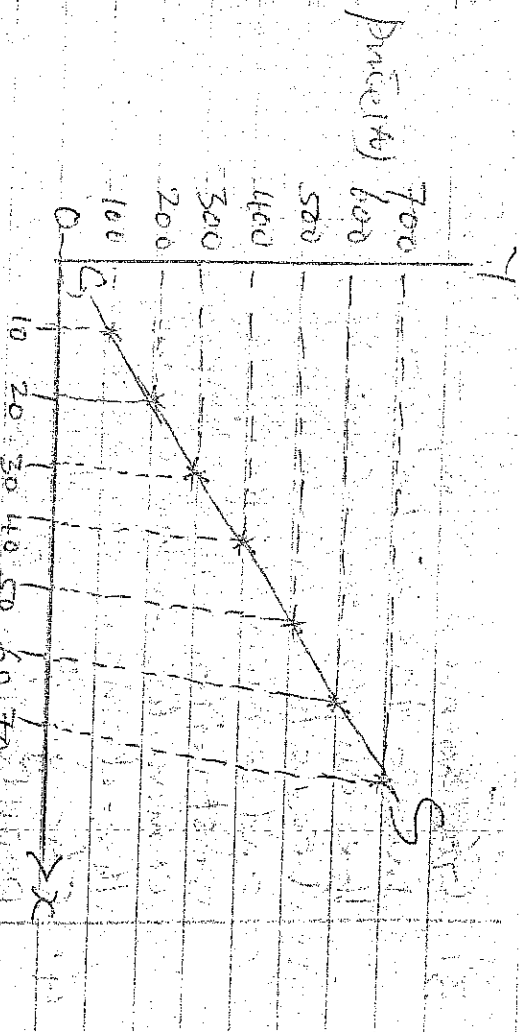


The downward sloping of the demand curve shows that more bread will be bought at lower price.

Supply Schedule - It shows the various quantities of commodity which sellers are willing to sell at various prices.

Price of bread (\$)	Qty supply
700	70
600	60
500	50
400	40
300	30
200	20
100	10
0	0

We can also represent the supply schedule graphically.



The supply curve slopes upward showing that more qty will be supplied at higher price.

DETERMINANTS OF DEMAND

There are several factors that determine the demand for a commodity.

1. Change in tastes and fashion. - Tastes relates to the likes of a consumer. Consumers would purchase more of commodity they prefer or like. If there is a change in taste in favour of a commodity the demand for it will be higher while a change in taste against a commodity will lead to lower demand. There is also a high demand for commodities which are in vogue while those out of fashion tends to be lower.

2. The price of substitutes - Goods with close substitute and complementary goods e.g. meat and fish, tea and coffee. If the price of one increases the consumer may switch over to the substitute. Thereby leading to a shift in the demand.

3. Government policy:- An increase in income tax by the government will reduce the disposable income of consumers which leads to decrease in demand for goods and services. Government may also place restriction on the importation of certain commodity in the country which reduces the demand for such commodity.

4. Change in population size:- If the population increases there is going to be an increase in the demand for food and other essentials of life.

5. New products:- Technological development brings new products to the market from time to time. The entry of new products reduces the demand for the old products etc.

DETERMINANTS OF SUPPLY

1. Cost of production:- If the cost of acquiring inputs goes up, the producer will demand less of these inputs. This means that the producer will produce less and therefore supply less of the commodity to the market. On the other hand, if the cost of production becomes cheaper, the producer will be willing to supply more.

2. Technological Development:- An improvement in the method of production may lead to an increase in supply.

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Natural Causes' - Drought, excessive rain, insect pest or crop diseases etc, destroys agricultural farms and affects the supply of such products. As Number of Producers' - Other things being equal the greater the total number of producers the greater the qty of goods supplied and vice versa. S: Government Policies' - Policies on subsidies and taxes on company may reduces the production of a commodity in the country. Increase excise duties,

EQUILIBRIUM PRICE AND QTY.

The equilibrium price is that price which equates demand and supply. It is determined by the interplay of the forces of demand and supply. In a free market economy, the equilibrium price is the market price for the commodity.

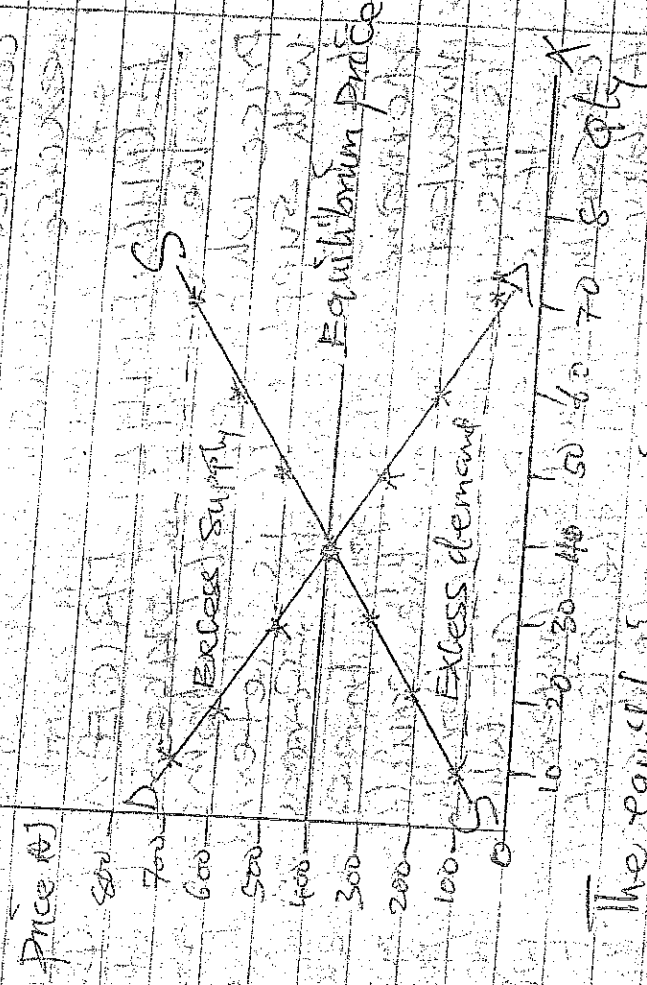
It is the price at which the qty of commodity which buyers are willing to buy equals the qty of that commodity which sellers are willing to offer for sale. Above the equilibrium price, more of the commodity will be supplied by seller than is demanded by consumers. Below the equilibrium price, more will be demanded by consumers than is supplied by producers or sellers. Equilibrium price can be illustrated by using demand and supply schedules as follows:

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Demand and Supply Schedule

Prices	Qty dd	Qty ss
700	10	70
600	20	60
500	30	50
400	40	40
300	50	30
200	60	20
100	70	10

We can represent this graphically



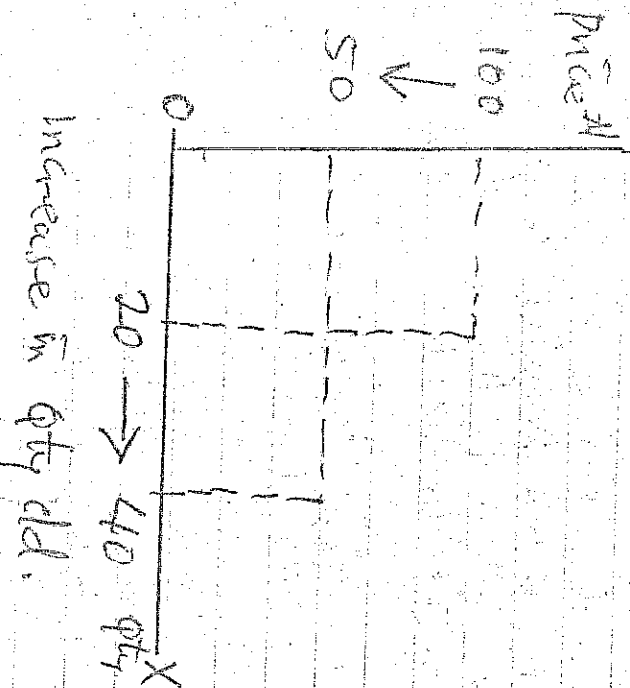
The equilibrium price is 400. At this price, 40 qty of the commodity are demanded and supplied at the same time. To clear the excess stock of goods (excess supply), producers or sellers has to cut the price! Cutting the price will increase the qty demand and reduces the qty producers wishes to supply.

MOVEMENT ALONG THE DEMAND CURVE AND SHIFT ALONG THE DEMAND CURVE.

Movement Along a Demand Curve (Change in Qty demanded).

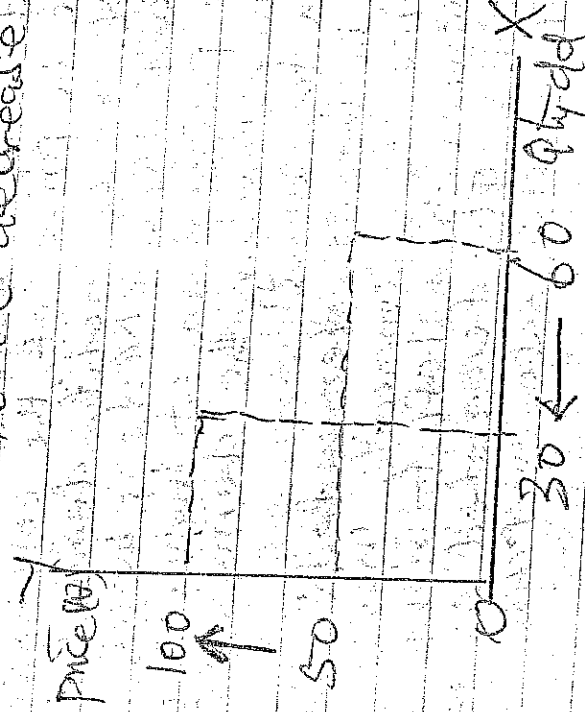
When there is a change in Qty demanded, the demand curve does not shift, rather there is a movement along a particular demand curve. So movement along a demand curve is caused by change in price of the commodity. The Qty of the commodity demanded changes with Price. More is bought at a lower price than at higher price.

1. Movement along the demand curve occur in two ways,
Increase in the Qty demanded. — Decrease in the price of a commodity moves the Qty demanded from left to right. Showing an increase in Qty demanded as the result of the decrease in price.



In the graph above, a decrease in the price of commodity from \$100 to \$50 brought about a move in the Qty ~~old~~ bought from 20 to 40 units.

ii) Decrease in Qty demanded: — There is going to be a decrease in the Qty demanded if the Qty bought decrease as a result of an increase in price. This cause the Qty demanded to move from right to left. i.e decrease in Qty. dd.



Increase in the price of the commodity from \$50 to \$100 brought about a decrease in Qty. dd. Hence, the Qty. dd move from right to left.

Shift in the Demand Curve (i.e. Change in Demand).

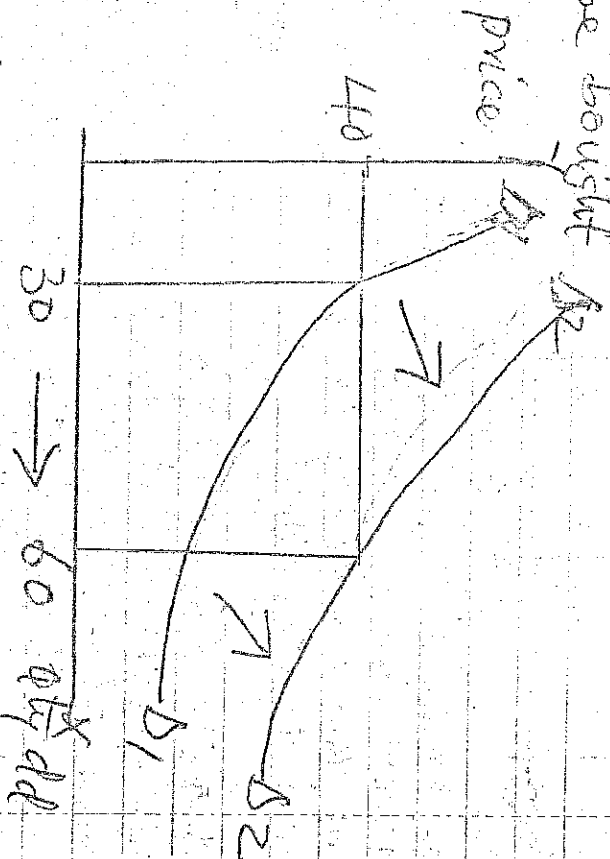
If the demand curve shift to an entirely new position, there is a change in demand. There is a completely new demand schedule and a demand

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curve, showing that at the old price, more or less of the commodity would be bought. A shift in demand curve is determined by the factors affecting demand other than the price of the commodity.

A shift in demand curve also occur in two ways,

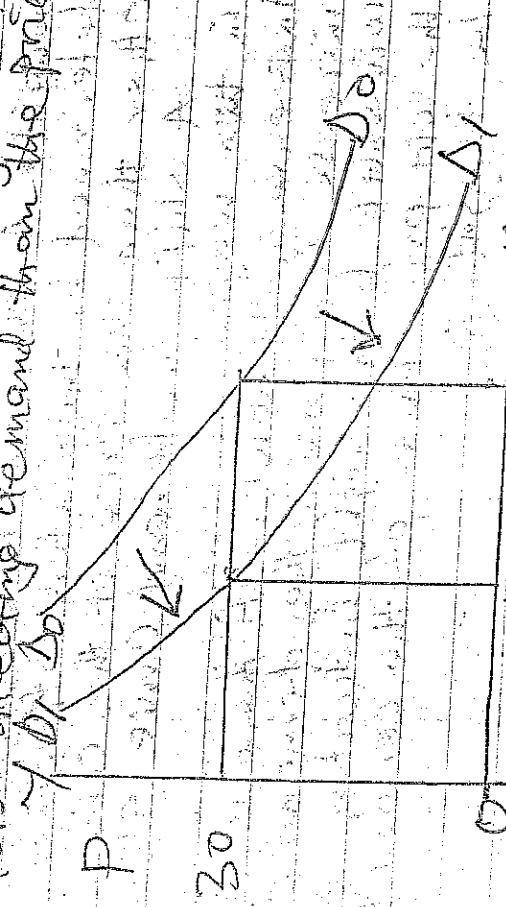
1. Increase in demand: - If there is an increase in demand, the demand curve will shift to the right, indicating that at the old price more of the commodity will be bought.



Here, the demand curve shifted from D_1 to D_2 at the old same price. Also, the Qty. d increased from 30 to 60 units.

ii) Decrease in demand: - If there is a decrease in demand, the demand curve shifts to the left, showing that at the old price less of the commodity is bought.

A decrease in demand is caused by an unfavourable change in any of the factors affecting demand than the price.



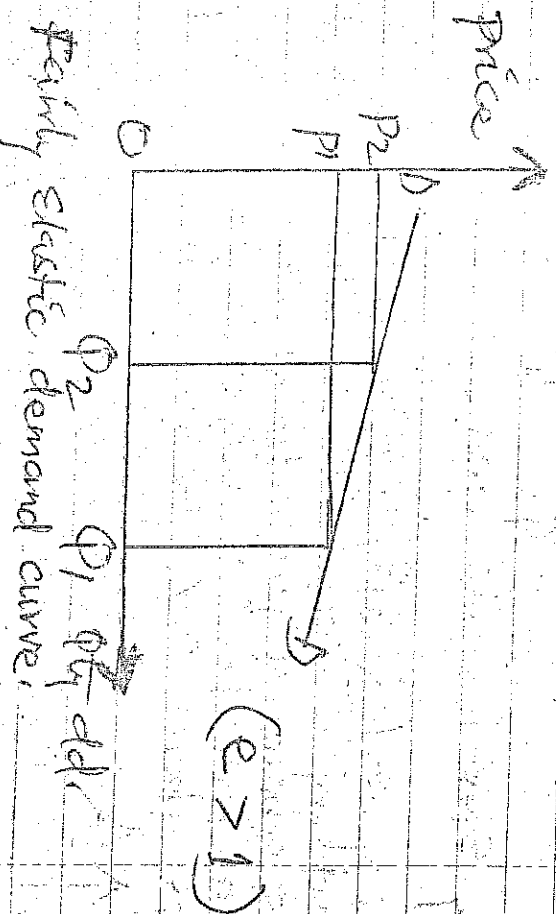
The demand curve shifted from 80 to 40 at the same price of 30. The qty of the commodity bought decrease from 80 units to 40 units.

ELASTICITY OF DEMANDS.

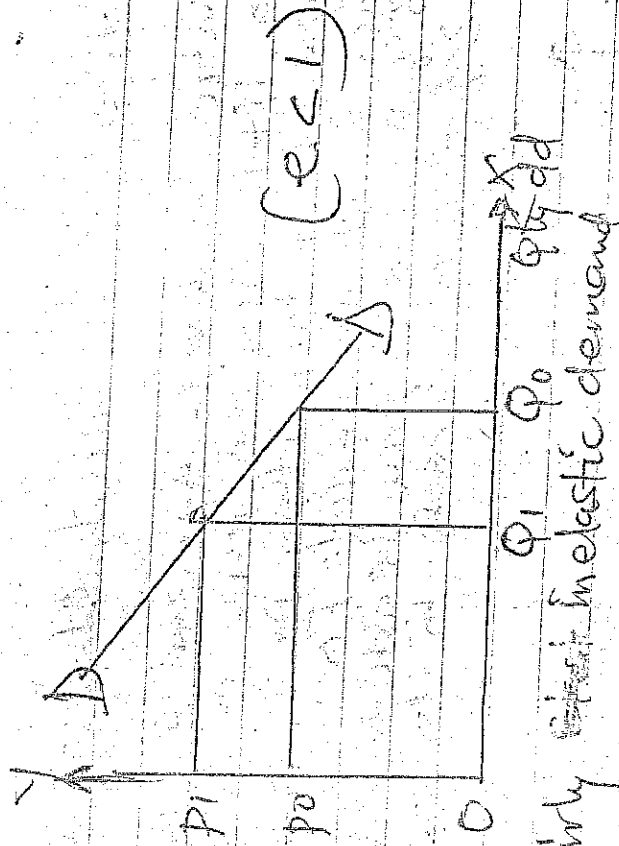
- Price elasticity of demand is the degree of sensitivity of demand (Qty of commodity) to a small change in price. Price elasticity of is often called elasticity of demand.
- Income elasticity of demand measures the degree of sensitivity of demand to a little change in income of the consumer.
- Cross elasticity of demand measures the degree of responsiveness of demand to a little change in the price of other commodity (ie complementary goods).

Types of Elasticity of Demand

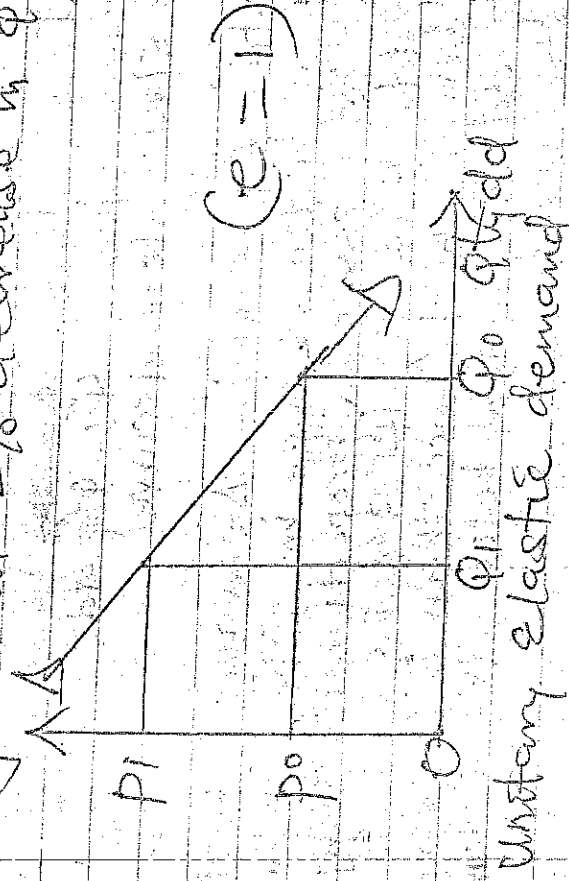
1. Elastic Demand - Demand is elastic if a little change in price or income brings about more than proportional change in qty demanded. For example if 3% increase in price leads to more than 3% fall or decrease in demand.



2. Inelastic Demand - Demand is inelastic if a little change in price or income leads to less than proportional change in qty d. Example, if 3% increase in price brings about less than 3% decrease in qty d. The graph below shows fairly inelastic demand.



3. Unitary Elasticity of Demand: - Demand is unitary elastic if a change in price brings about a proportional change in qty demanded. If 2% increase in price brings about 2% decrease in qty dd.



4. Perfectly Elastic demand: - Demand is perfectly elastic if a slight increase in price will make the consumers to stop buying the commodity at all. While a

Slight decrease in price will make the consumers to purchase all the qty of that commodity. The coefficient of elasticity of demand is infinite ($e = \infty$).



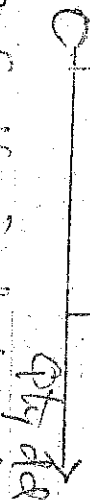
Perfectly elastic demand

S, 'Perfectly inelastic' - Demand is perfectly inelastic if any change in price does not make the qty demanded to change. The demand curve is parallel to the price. The same qty is demanded irrespective of changes in price.

Price $\downarrow$



$$e = 0$$



Perfectly inelastic demand

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MEASUREMENT OF PRICE ELASTICITY OF DEMAND.

Elasticity of demand can be measured or determined by calculating elasticity of demand coefficient.

Formula: Elasticity of demand or price elasticity of demand =

$$\frac{\% \text{ change in qty demanded}}{\% \text{ change in price}}$$

Income Elasticity of demand = $\frac{\% \text{ change in qty dd}}{\% \text{ change in income}}$

Cross elasticity of demand = $\frac{\% \text{ change in commodity X}}{\% \text{ change in commodity Y}}$

$\frac{\% \text{ change in the price of commodity Y}}{\% \text{ change in the price of commodity X}}$

1. Numerical Example:

If the price of bread increased from ₹ 25 to ₹ 35 and the qty bought per week decrease from 300 to 200 loaves determine the elasticity of demand for bread.

Solution

Original qty dd = 300 loaves

New qty dd = 200 "

Change in qty dd = $300 - 200 = 100$

$$\% \text{ change in qty dd} = \frac{100}{300} \times 100 = \frac{100}{3} = 33\frac{1}{3}\%$$

Original price of bread = ₹ 25

New price of bread = ₹ 35

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$$\text{Change in Price} = 1035 - 1025 = 10$$

$$\% \text{ change in price} = \frac{10}{25} \times 100 = 40\%$$

$$\text{Price elasticity of demand} = \frac{\% \text{ change in qty. dmd}}{\% \text{ change in price}}$$

$$= \frac{33\frac{1}{3}\%}{40\%} = \frac{100}{3} = \frac{100 \times 100}{3 \times 40}$$

$$= \frac{100}{3} \times \frac{1}{40} = \frac{25}{30} = \frac{5}{6} = 0.83$$

The elasticity of demand is less than one, the demand for bread is inelastic.

2. If 5% increase in price leads to 10% fall in qty demand, calculate the elasticity of demand.
Solution

$$\text{Price elasticity of demand} = \frac{\% \text{ change in qty. dmd}}{\% \text{ change in price}}$$

$$= \frac{10\%}{5\%} = \frac{10}{100} \div \frac{5}{100} = \frac{10}{100} \times \frac{100}{5}$$

$$= \frac{10}{5} = 2 \quad \text{Since elasticity of}$$

demand is greater than one, demand is fairly elastic.

THE CONSUMER BEHAVIOUR

The concept allow us to predict how consumers will respond to changes in market conditions.

The concept of utility:- Utility in economics refers to the amount of satisfaction that a consumer derived from the consumption of a commodity at a particular time. Utility is related to time - the amount of utility derived from a particular commodity may not be the same for the same individual at different times.

Cardinal and Ordinal utility

Cardinal utility:- The Cardinal School of thought are of the view that utility of a commodity can be measured subjectively in units called 'utils'. Others suggested that utility can be measured in monetary units by relating it to the amount of money which the consumer is willing to pay for a given quantity of commodity at a particular time.

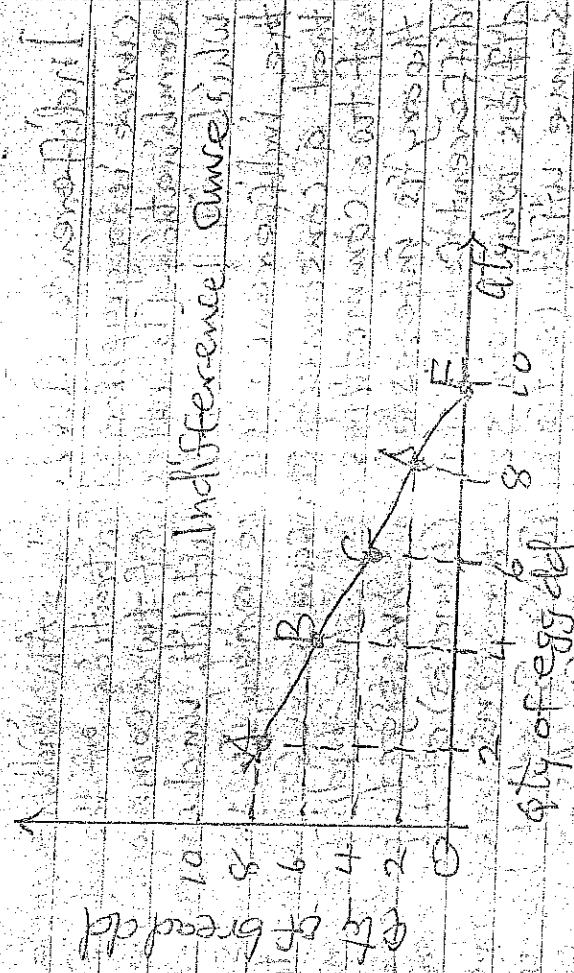
Ordinal utility:- Economists who belong to this school argued that it is not possible to measure utility. They opine that although utility cannot be precisely measured, it is possible for a consumer to make a choice between various bundles of commodities by ranking them according to the level of satisfaction expected from each bundle without specifying exact units of utility.

Indifference Curve - An indifference curve is a graphical illustration of the different combinations (bundles) of two commodities which will leave total utility unchanged. In the indifference curve analysis, we assume that a consumer consumes a combination of two commodities only. The indifference theory is interested in investigating the different combinations (bundles) of two commodities which will give the consumer the same utility (satisfaction). So that he is indifferent in choosing any combination of the two commodities which provide same satisfaction.

For example, consider two commodities, egg and bread as shown below:

Combinations	Egg Qty	Bread Qty	Total Utility
A	2	8	10
B	4	6	10
C	6	4	10
D	8	2	10
E	10	0	10

Now we can use the combinations of the two commodities (egg and bread) to illustrate graphically the indifference curve.



Total utility is equal at any point on the graph. The consumer gets the same level of satisfaction. The consumer may choose combination A, 8 slice of bread and 0 eggs or combination C, 4 slice of bread and 4 eggs, all gives the same level of satisfaction. The consumer can make his choice between these combinations of two commodities by ranking them according to the level of satisfaction expected from each bundle.

Marginal Rate of Substitution

Marginal rate of substitution is the qty of one of the two commodities that the consumer must sacrifice to increase the qty of other commodity without changing the total utility. In our example above, the marginal rate of substitution of eggs for bread is the quantity of bread the consumer must sacrifice to increase the qty of eggs by one without changing total utility.

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